A ZERO-KNOWLEDGE DEMO · BLOCKCHAIN CLUB

Battleship, proven.

A zk-SNARK demo with real on-chain verification.

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noir → ultrahonk → solidity

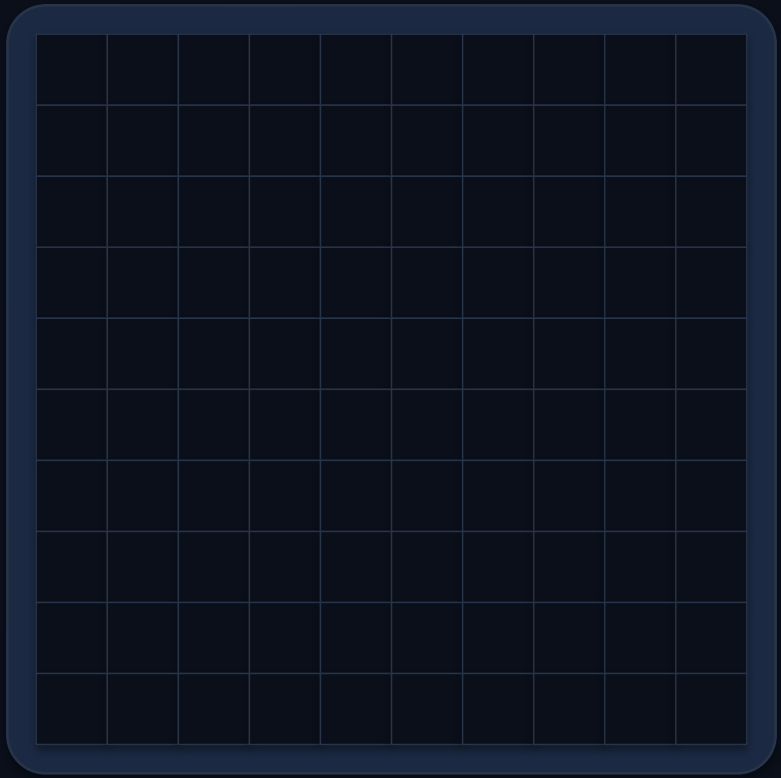
A game built on hidden state.

> **Each player knows their board.**
> **Neither player trusts the other.**

— standard trick: commit to your board with a hash so you can't change your mind later.

But a hash commitment only proves WHAT you committed to -- not whether what you committed is legal.

Commit to nothing. Win anyway.



ALL-EMPTY BOARD

hash(empty, salt) is still valid

THE CHEAT

1. **Commit to an all-empty board.**
   hash is valid. nothing in the scheme says you need ships.
2. **Opponent fires at (3, 5).**
   you answer MISS. truthfully. there is nothing there.
3. **Every shot is a miss.**
   you never lose a cell. you never take damage.
4. **You win by elimination.**
   hash commitments alone cannot stop this.

Prove legality, in zero knowledge.

The proof IS the validity certificate.

One zk-SNARK, generated in the browser before the first shot.

Verified once on-chain. No ship-count lies survive.

THE FLEET CONSTRAINT

17 cells - 1x5 - 1x4 - 2x3 - 1x2 - no overlaps - no diagonals

Noir + UltraHonk.

NOIR

Rust-like zk DSL

Readable circuits.
Compiles to arithmetic constraints.
WASM prover runs in the browser.

ULTRAHONK

PLONK-family backend

Aztec's modern proof system.
No trusted setup ceremony.
No powers-of-tau, no toxic waste.

PEDERSEN HASH

In-circuit commitment

Cheap to compute inside zk.
Binds 100 cells + salt.
One field element, used as the board commit.

HONKVERIFIER

Auto-generated Solidity

~2,460 lines of verifier.
~250k gas per verify call.
Dropped straight into the repo.

Browser proves. Chain verifies.

FLEET
Fleet
private[100]

→

PED
Pedersen
hash → commit

→

NOIR
Noir circuit
board_validity

→

HONK
UltraHonk
proof blob

→

SOL
HonkVerifier
Solidity verify

→

GAME
BattleshipGame
settle on-chain

BROWSER → WASM PROVER → NOIR → SOLIDITY VERIFIER → CHAIN

- Browser proves, chain verifies
- Burner wallets sign, zero popups
- Local Anvil for instant demo

LIVE DEMO →

http://localhost:5173

two burner wallets · one laptop · zero trust

What was proven on-chain.

1. **Board legality before the first shot**

   Can't commit an all-empty board. The circuit rejects it.

2. **Every shot response consistent with the committed board**

   Hit or miss, bound back to the original Pedersen commitment.

3. **Zero trust in the opponent's claims**

   Every answer is a proof the contract verifies, not a promise.

4. **Zero trusted-setup ceremony**

   UltraHonk means no multi-party ritual, no toxic waste.

What's next.

01 **Mainnet deployment path**

Solidity verifier is deploy-ready.

02 **Circuit optimization**

Shrink shot-response constraints.

03 **Multi-game lobby**

Matchmaking plus state routing.

Q & A

Questions?

Circuits, contracts, frontend -- everything is open source.

github.com/ battleship-zk